The generalization of the non-relativistic Luttinger-Zeeman action to four spatial dimensions

Santanu Dey
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We derive the non-relativistic Luttinger-Zeeman action in arbitrary dimensions.

I. INTRODUCTION

Let's consider the $j = 3/2$ spin-orbit coupled Luttinger Hamiltonian in the presence of an external magnetic field in $d = 3$. In the interest of a higher dimensional extension, we write down a Hamiltonian

$$\begin{aligned} H &= \frac{4}{2(d-1)}\left[\left(\vec{p}\cdot\vec{S}\right)^2 - \frac{j(j+1)}{d}p^2\right] + g\vec{B}\cdot\vec{S}, \text{ with,} \\ [S_i, S_j] &= i\epsilon_{ijl}S_k \text{ for } d = 3. \end{aligned} \tag{1}$$

The Hamiltonian acts on $j = 3/2$ fermionic spinors. SO(3) rotation of the three-dimensional Euclidean space determines the form of the Zeeman coupling in the Hamiltonian. In a field-theoretic language, the momentum-space Lagrangian corresponding to this non-relativistic Hamiltonian can be written as

$$\mathcal{L} = \psi^\dagger\left[\partial_\tau + H\right]\psi, \tag{2}$$

where ψ is a four-component SU(2) spinor with $j = 3/2$. Under an infinitesimal SO(3) rotation, we have the following transformation:

$$\substack{R\in\mathrm{SO}(3),\\ U(R)\in\mathrm{SU}(2)} : \begin{cases} B_i \mapsto R_{ij}B_j, \\ \psi^\dagger S_i\psi \mapsto \psi^\dagger U^\dagger(R)S_iU(R)\psi = R_{ij}\psi^\dagger S_j\psi \end{cases} \tag{3}$$

This demonstrates that the Zeeman vertex $\vec{B}\cdot\vec{S}$ is rotationally invariant. That the $j = 3/2$ spin matrices $\vec{S}$ also transform like vector under SU(2) transformation U follows from the Lie-algebraic isomorphism between the two groups

$$\mathrm{Lie[SU(2)]} = \mathrm{Lie[SO(3)]}, \tag{4}$$

and it can be proved at the level of infinitesimal transformation by exploiting the algebraic structures of the two groups.

The extension of the Zeeman coupling to higher dimension d hinges on the isomorphism between the Lie algebras of the rotation group SO(d) and the corresponding complex spinorial group Spin(d). For $d = 3$, we have, Lie[Spin(3)] = Lie[SU(2)] = Lie[SO(3)]. For $d = 4$, however, SO(4) has 6 generators with a Lie algebra,

$$\begin{aligned} [J_i, J_j] &= i\epsilon_{ijk}J_k, \\ [J_i, K_k] &= i\epsilon_{ijk}K_k, \\ [K_i, K_j] &= i\epsilon_{ijk}J_k, \end{aligned} \tag{5}$$

which can be decoupled with the parametrization $J_i^1 = (J_i + K_i)/2$ and $J_i^2 = (J_i - K_i)/2$,

$$\begin{aligned}[J_i^1, J_j^1] &= i\epsilon_{ijk} J_k^1, \\ [J_i^2, J_j^2] &= i\epsilon_{ijk} J_k^2, \\ [J_i^1, J_j^2] &= 0,\end{aligned} \tag{6}$$

into two SU(2) Lie algebras. In other words [1],

$$\text{Lie[SO(4)]} = \text{Lie[Spin(4)]} = \text{Lie[SU(2)]} \oplus \text{Lie[SU(2)]}. \tag{7}$$

The spinor representation of SO(4) is therefore characterized by the j quantum numbers of the two SU(2) groups, labeled by their spins (j_1, j_2) with a combined highest weight $j_1 + j_2$. First, we need to define spinors with total spin $j = 3/2$ in the four dimensional space. This can be accomplished by choosing between $(3/2, 0)$, $(1, 1/2)$, $(1/2, 1)$, and $(0, 3/2)$ representations. By choosing one of the two half-integer, fully-fermionic representations,

$$\begin{aligned}(\frac{3}{2}, 0) &: \vec{J}_1 = \vec{S}, \vec{J}_2 = 0, \\ (0, \frac{3}{2}) &: \vec{J}_1 = 0, \vec{J}_2 = \vec{S},\end{aligned} \tag{8}$$

we can define four-component $j = 3/2$ spinor fields. But this is not enough to obtain a rotationally invariant theory of fermionic spinors coupled together with bosonic vector fields. The spin one bosonic fields B_i transforms as SO(4) vectors with the six generators of rotation $[\mathcal{J}^{ij}]_{kl} = i(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk})$,

$$B_k \mapsto \left(\delta_{kl} - \frac{i}{2}\theta_{ij}[\mathcal{J}^{ij}]_{kl}\right) B_l. \tag{9}$$

with infinitesimal rotation generated by the antisymmetric matrix θ. The six rotation generators can be compactly combined as,

$$\mathcal{J} = \begin{pmatrix} 0 & K_1 & K_2 & K_3 \\ -K_1 & 0 & J_3 & -J_2 \\ -K_2 & -J_3 & 0 & J_1 \\ -K_3 & J_2 & -J_1 & 0 \end{pmatrix}, \tag{10}$$

with the generators J_i and K_i following the algebra mentioned above. In the (j_1, j_2) spinor basis, they are respectively spin $j_1 + j_2$ and spin $j_1 - j_2$ matrices. Evidently, in a $(j, 0)$ spinor basis, distinct SO(4) rotations generated by, e.g., $\mathcal{J}_{12}$ and $\mathcal{J}_{34}$ leads to the same transformation on the spinor field, as the latter transforms under the superposed SU(2) generator $\vec{J}_1 = (\vec{J} + \vec{K})/2$. We shall, therefore, need Dirac spinors in the $(3/2, 0) \oplus (0, 3/2)$ representation that transform uniquely under rotation.

Now, we have a $2 \times (2 \times 3/2 + 1) = 8$ dimensional spinor representation in a four dimensional space, and we wish write down an invariant scalar involving the spinor and tensor fields. For ordinary Dirac spinors in $(1/2, 0) \oplus (0, 1/2)$ representation, this is achieved through generators of the Euclidean Clifford algebra in the spinor space,

$$\begin{aligned}\left\{\Gamma^i, \Gamma^j\right\} &= 2\delta^{ij}, \\ \Gamma^{ij} = \frac{i}{2}\left[\Gamma^i, \Gamma^j\right], S^{ij} &= \frac{1}{2}\Gamma^{ij}.\end{aligned} \tag{11}$$

The matrices S^{ij} generates SO(d) rotation

$$\begin{aligned}[\Gamma^i, S^{kl}] &= [\mathcal{J}^{kl}]^i{}_j \Gamma^j, \\ \left[S^{ij}, S^{kl}\right] &= i\left(\delta^{jk}S^{il} - \delta^{ik}S^{jl} - \delta^{jl}S^{ik} + \delta^{il}S^{jk}\right).\end{aligned} \tag{12}$$

Note, that in the Euclidean space, raising and lowering indices do not introduce any sign factors. Using the first commutation relation, it is now straightforward to show that under unitary transformation $U = \exp\left(-(i/2)\theta_{ij}S^{ij}\right)$ the Clifford matrices transform as,

$$U^\dagger \Gamma^i U = \exp\left(-\frac{i}{2}\theta_{kl}\mathcal{J}^{kl}\right)^i_{\ j} \Gamma^j = R^i{}_j \Gamma^j. \tag{13}$$

This implies that under $SO(d)$ rotation generated by $d(d-1)/2$ distinct Γ^{ij} matrices, the $(1/2, 1/2)$ spinor bilinear product $\psi^\dagger \Gamma^i \psi$ transform as vectors. Similarly, the product $\psi^\dagger \Gamma^{ij} \psi$ transform as rank-2 tensors. And a Zeeman coupling can be expressed as the invariant product

$$\psi^\dagger \Gamma^{ij} \psi F_{ij}. \tag{14}$$

In $d = 2$ and 3, the product reduces to the familiar Zeeman coupling.

Now, dimensionality of the spinor space also restrict the number of generators of the Clifford group. For a 2-dimensional (Weyl) spinor in the representation $(1/2, 0)$, there are three 2×2 generators given by the Pauli matrices

$$\Gamma^1 = \sigma^1, \Gamma^2 = \sigma^2, \Gamma^3 = -i\sigma^1\sigma^2 = \sigma^3, \tag{15}$$

with Γ^{ij} generating SO(3) rotation. For the 4-dimensional $(1/2, 0) \oplus (0, 1/2)$ Dirac representation, five 4×4 Clifford matrices can be constructed by doing a Kronecker product of the 2×2 Clifford matrices with σ^1 and adding a component $\sigma^2 \otimes 1$ [2],

$$\begin{aligned}&\Gamma^1 = \sigma^1 \otimes \sigma^1, \Gamma^2 = \sigma^1 \otimes \sigma^2, \\ &\Gamma^3 = \sigma^1 \otimes \sigma^3, \Gamma^4 = \sigma^2 \otimes 1, \\ &\Gamma^5 = (-i)^2 \Gamma^1\Gamma^2\Gamma^3\Gamma^4 = \sigma^3 \otimes 1,\end{aligned} \tag{16}$$

that constitutes an SO(5) Clifford algebra. The last Clifford matrix is obtain by, $\Gamma^{2n+1} = (-i)^n \Gamma^1\Gamma^2 \cdots \Gamma^{2n}$. The construction reveals that $2n + 1$ Clifford matrices are needed to span a spinor space of 2^n dimension.

For our $2^3 = 8$ dimensional $(3/2, 0) \oplus (0, 3/2)$ Dirac spinors, we shall need a construction with 7 Clifford matrices. The construction may used to generate SO(7) rotation or two distinct SO(6) rotations with a (chiral) projection operator $P_\pm = (1 \pm \Gamma^7)/2$ acting on the spinor space, splitting the 8 dimensional Dirac representation into two irreducible 4 dimensional Weyl representation $(3/2, 0)$ and $(0, 3/2)$. By sticking to the Dirac representation, a SO(7) symmetric scalar can be written down in the eight-dimensional space Z^a,

$$\psi^\dagger \Gamma^{ab} \psi F_{ab}. \tag{17}$$

Now, a reduction in dimensionality can be performed by declaring that $F_{ab}(Z) = F_{ab}(x)$, where x^i is the four-dimensional space of our interest. By the virtue of $F_{ab}(x) = \partial_{Z_a} A_b(x) - \partial_{Z_b} A_a(x)$ being the field strength tensor, the Zeeman coupling is uniquely determined.

II. SO(9) CLIFFORD ALGEBRA AND HOW MURAKAMI ET AL. [3] WRITES THE ZEEMAN TERM

Following Murakami et al. [3] and Janssen et al.'s [4] parametrization, the Clifford matrices are defined as,

$$\Gamma^a = \sqrt{\frac{1}{2d(d-1)}}\Lambda^a_{ij}\{S^i, S^j\} \text{ with } \Lambda^a_{ij}\Lambda^a_{kl} = \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \frac{2}{d}\delta_{ij}\delta_{kl}, \tag{18}$$

where the second formula is due to Janssen et al. who consider a d dimensional generalization of the problem from $d = 3$. S_i are the spin $j/2$ angular-momentum matrices satisfying the SU(2) algebra. Now, let us introduce the functions, $d_a(\vec{p}) = \sqrt{\frac{d}{2(d-1)}}\Lambda^{ij}_a p_i p_j$. From the above property of the Λ^a matrices, it follows that,

$$\begin{aligned}
\Gamma^a d_a(\vec{p}) &= \frac{1}{2(d-1)}\Lambda^a_{ij}\Lambda^{kl}_a\{S_i, S_j\}p_k p_l, \\
&= \frac{4}{2(d-1)}\left[\left(\vec{p}\cdot\vec{S}\right)^2 - \frac{j(j+1)}{d}p^2\right], \\
&= H
\end{aligned} \tag{19}$$

We are interested in enumerating the rotation generators $\Gamma^{ab} = \frac{i}{2}[\Gamma^a, \Gamma^b]$ and their relation to the $j = 3/2$ spin operator $S^i = \eta^i_{ab}\Gamma^{ab}$. We invert the relation,

$$\begin{aligned}
&\text{tr}[S^i\Gamma^{cd}] = \eta^i_{ab}\text{tr}[\Gamma^{ab}, \Gamma^{cd}], \\
&\text{tr}[S^i\Gamma^{cd}] = -d_\Gamma\eta^i_{ab}\left(\delta^{ad}\delta^{bc} - \delta^{ac}\delta^{bd}\right), \\
&\eta^{cd}_i = \frac{1}{2d_\Gamma}\text{tr}[S_i, \Gamma^{cd}],
\end{aligned} \tag{20}$$

where d_Γ is the matrix dimension of the Clifford matrices and we have used the antisymmetry properties of the η^i_{ab} symbol to arrive at the result. Now, by substituting the definition of Γ^a matrices in terms of the spin matrices we have,

$$\eta^{ab}_i = \frac{i}{2d_\Gamma}\frac{1}{2d(d-1)}\Lambda^a_{mn}\Lambda^b_{rs}\text{tr}\left[S^i[\{S^m, S^n\}, \{S^r, S^s\}]\right]. \tag{21}$$

The Zeeman coupling can now be written down as,

$$\begin{aligned}
&\psi^\dagger S^i B_i\psi, \\
=&\psi^\dagger\Gamma^{ab}\psi\frac{\eta^{ab}_i\epsilon_{ijk}}{2}F^{jk}, \\
=&g^{ab}_{jk}\psi^\dagger\Gamma^{ab}\psi F^{jk}.
\end{aligned} \tag{22}$$

In the current computation, it is this g^{ab}_{jk} symbol that I computed using the d dimensional formulae for Λ^a_{ij} and δ_{ij}.

III. JANSSEN ET AL.'S [4] WAY OF DOING HIGHER DIMENSIONAL EXTENSION

With $(d+2)(d-1)/2$ number of Γ^a matrices, Janssen et al. writes down a general quadratic band touching (QBT) Hamiltonian,

$$H = G_{ij}p_ip_j \text{ with } G_{ij} = \sqrt{\frac{d}{2(d-1)}}\Lambda^a_{ij}\Gamma_a, \tag{23}$$

in general d dimensions. From the earlier discussion, we can surmise that in $d = 4$, the nine 16×16 Clifford matrices can be combined to construct a sixteen-dimensional SO(9) representation. Now, let us think about the Fermioic spinor space for a moment. We have moved from three-dimensional Euclidean space with $SO(3)$ rotational invariance to four-dimensional Euclidean space with $SO(4)$ rotational symmetry. Furthermore, we wish to have a quantized description of the fermionic fields in their spinor representation. In the three-dimensional space, the label j of the SU(2) spinors describe the spin of the homomorphic representation SO(3). With $\mathrm{SO}(4) = \mathrm{SU}(2) \otimes \mathrm{SU}(2)/\mathbb{Z}_2$, the notion of spin is associated with the angular momentum of its SO(3) subalgebra. The decomposition goes as follows (see Sec. I for explanation):

Dimensions	2	4	4	8	16	16
Rep. of $\mathrm{SU}(2) \otimes \mathrm{SU}(2)$	(1/2,0)	(1/2,1/2)	$(1/2, 0) \oplus (0, 1/2)$	$(3/2, 0) \oplus (0, 3/2)$	(15/2,0)	$(7/2, 0) \oplus (0, 7/2)$
Rep. of SO(3) sub-algebra	1/2	$1 \oplus 0$	1/2	3/2	15/2	7/2

Clearly, a half-integer spin description of the QBT Hamiltonian involves fermions of $j > 3/2$ angular momentum, with the identification of the angular momentum operators as the generator of the SO(3) subalgebra of the SO(4) spatial symmetry. However, this characterization holds onto the canonical notion of angular momentum and spin tied to a SO(3) submanifold of SO(4). In principle, we could dispense with this construction and notions of subspace spins altogether. We could instead identify combinations of the six generators $\vec{J}^1$ and $\vec{J}^2$ of a (j_1, j_2) representation of $\mathrm{SU}(2) \otimes \mathrm{SU}(2)$ as the six generators $\vec{J} = (\vec{J}_1 + \vec{J}_2)/2$, $\vec{K} = (\vec{J}_1 - \vec{J}_2)/2$ of the SO(4) rotation. In this way, the sixteen-dimensional fermionic spinor space can be described with a $(j_1, j_2) = (3/2, 3/2)$ Weyl representation (see Sec. I for the notation used). This is inherently a multi-integer spin representation of the SO(4) rotation group,

$$(3/2, 3/2) = 3 \oplus 2 \oplus 1 \oplus 0 \oplus 0, \tag{24}$$

where the last zero comes from the $\vec{K}$ representation of SO(4)/SO(3). However, these are artificial spinors aiding us towards a dimensional increment of the three-dimensional $j = 3/2$ QBT fermions. We wish to write down a Zeeman like coupling in four-dimensional space that reduces to a regular three-dimensional Zeeman term after dimensional reduction. The obvious candidate is the term,

$$H_Z = \psi^\dagger \Sigma_{ij} \psi F_{ij}, \tag{25}$$

where F_{ij} and Σ_{ij} are second rank tensors under SO(4) rotation to form this invariant coupling. Under the rotation of the SO(4) vector space, we must have,

$$\substack{R \in \mathrm{SO}(4), \\ U(R) \in \mathrm{SU}(2) \otimes \mathrm{SU}(2)} : \begin{cases} F_{ij} \mapsto R_{ik}R_{jl}F_{kl}, \\ \psi^\dagger \Sigma_{ij}\psi \mapsto \psi^\dagger U^\dagger(R)\Sigma_{ij}U(R)\psi = R_{ik}R_{jl}\psi^\dagger \Sigma_{kl}\psi \end{cases} \tag{26}$$

SO(4) spatial rotations in Janssen et al.'s construction with a large sixteen-dimensional spinor space leaves the quadratic band touching tensor G_{ij} invariant. For an infinitesimal rotation, $R_{ij} = \delta_{ij} - (i/2)\theta_{kl}[\mathcal{J}^{kl}]_{ij}$ and $U_{\alpha\beta} = \delta_{\alpha\beta} - (i/2)\theta_{kl}[S^{kl}]_{\alpha\beta}$, the construction yields,

$$\begin{aligned} U^{\dagger}(R)G_{ij}U(R) &= R_{ik}R_{jl}G_{kl} \\ G_{ij} - \frac{i}{2}\theta_{uv}[G_{ij}, S^{uv}] &= G_{ij} - \frac{i}{2}\theta_{uv}\left([\mathcal{J}^{uv}]_{ik}\delta_{jl} + \delta_{ik}[\mathcal{J}^{uv}]_{jl}\right)G_{kl}. \end{aligned} \tag{27}$$

The sixteen-dimensional spinor space is spanned by the matrices 1, Γ^a, and Γ^{ab}, so we can decompose $S^{uv} = A^{uv} + B^{uv}_a\Gamma^a + C^{uv}_{ab}\Gamma^{ab}$. In order to derive these A, B, and C symbols, we compute the commutator in the left hand side and find that,

$$[G_{ij}, S^{uv}] = -(2\sqrt{2}i)\sqrt{\frac{d}{d-1}}\left[C^{uv}_{ab}\Lambda^b_{ij}\Gamma^a + B^{uv}_a\Lambda^{ij}_e\Gamma^{ea}\right]. \tag{28}$$

It is clear that for S^{uv} to generate SO(4) rotation, the symbols B^{uv}_a must identically vanish. For the right hand side, we have the expression,

$$\begin{aligned} &\left([\mathcal{J}^{uv}]_{ik}\delta_{jl} + \delta_{ik}[\mathcal{J}^{uv}]_{jl}\right)G_{kl} \\ =& -i\sqrt{\frac{d}{2(d-1)}}\Gamma^e\left(\delta_{iv}\Lambda^e_{ju} - \delta_{iu}\Lambda^e_{vj} + \Lambda^e_{iu}\delta_{jv} - \Lambda^e_{iv}\delta_{ju}\right). \end{aligned} \tag{29}$$

Comparing both sides we have the relation for the C symbols,

$$4C^{uv}_{ab}\Lambda^b_{ij} = \left(\delta_{iv}\Lambda^a_{ju} - \delta_{iu}\Lambda^a_{vj} + \Lambda^a_{iu}\delta_{jv} - \Lambda^a_{iv}\delta_{ju}\right), \tag{30}$$

from which the antisymmetry of the symbol under $u \leftrightarrow v$ is evident, as expected. To enumerate the symbol, we contract both sides with the Λ^c_{ji}, and use the property $\mathrm{tr}\left[\Lambda^a\Lambda^b\right] = 2\delta^{ab}$ [4] to obtain,

$$C^{uv}_{ab} = \frac{1}{4}\left(\Lambda^a_{ui}\Lambda^b_{iv} - \Lambda^a_{vi}\Lambda^b_{iu}\right) = \frac{1}{4}[\Lambda^a, \Lambda^b]_{uv}, \tag{31}$$

which completes our mapping of the SO(4) rotation generator in this higher-dimensional spinor space

$$\begin{aligned} S^{ij} &= C^{ij}_{ab}\Gamma^{ab}, \\ &= \frac{i}{4}\left(\left[\Gamma^a\Lambda^{ik}_a\right]\left[\Gamma^a\Lambda^{kj}_a\right] - \left[\Gamma^a\Lambda^{jk}_a\right]\left[\Gamma^a\Lambda^{ki}_a\right]\right) \end{aligned} \tag{32}$$

In an analogy to the lower-dimensional Zeeman coupling (the spatial part of the Pauli term of a spatio-temporal electrodynamics) we now write down an expression for a higher-dimensional Zeeman Hamiltonian,

$$H_Z = \frac{1}{2}\psi^{\dagger}\Gamma^{ab}\psi C^{ij}_{ab}F_{ij} \tag{33}$$

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[1] As exponentiation of Lie algebras generate Lie groups we can say $\mathrm{SU}(2)\otimes\mathrm{SU}(2) \to \mathrm{SO}(4)$. In fact, this is a double cover, $\mathrm{SO}(4) \simeq \mathrm{SU}(2)\otimes\mathrm{SU}(2)/\mathbb{Z}_2$.

[2] P. Ramond, *Group Theory: A Physicist's Survey* (Cambridge University Press, 2010).
[3] S. Murakami, N. Nagosa, and S.-C. Zhang, SU(2) non-abelian holonomy and dissipationless spin current in semiconductors, Phys. Rev. B **69**, 235206 (2004).
[4] L. Janssen and I. F. Herbut, Nematic quantum criticality in three-dimensional fermi system with quadratic band touching, Phys. Rev. B **92**, 045117 (2015).